Roee Tekoah

Security Researcher

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Security researcher with **5+ years of technical experience** across security research, systems, algorithms, and scientific software development. Currently completing a B.Sc. in Computer Science at the Hebrew University of Jerusalem (expected 2027).

EXPERIENCE

- **JSOF** · Security Researcher · Apr 2022–Jan 2024

  Black-box vulnerability research and attack-surface analysis of automotive and embedded platforms for intelligence agencies and commercial clients

  IDA Pro / IDAPython · ARM · Android · Automotive ECUs · Black-box RE

  - **ECU platform.** Analyzed a complex automotive ECU platform, reverse engineering Android, custom-architecture, and proprietary-OS subsystems from partial and corrupted memory dumps
  - **Vulnerability research.** Investigated automotive and embedded systems for potential vulnerabilities through black-box analysis, attack-surface assessment, and reverse engineering
  - **Tooling.** Built custom IDAPython tooling and reverse-engineering workflows to analyze obfuscated binaries and investigate potential attack paths in black-box environments

- **IDF (Matzov)** · Malware Researcher · Apr 2019–Apr 2022

  Large-scale malware research, collection, and detection for intelligence and defense teams

  C/C++ · Go · .NET · Python · Shellcode · YARA / Snort / Sigma · Memory forensics

  - **Malware analysis.** Reverse engineered obfuscated, multi-stage malware using static and dynamic analysis, memory forensics, and dump analysis, recovering payloads and behavior hidden behind anti-analysis defenses
  - **Live-sample hub.** Expanded sample and IOC coverage through manual extraction, honeypot integration, and pipeline refinement as part of a national-scale intelligence collection effort
  - **Detection and attribution.** Produced IOCs, YARA/Snort/Sigma signatures, and threat reports; developed internal tooling and automated workflows for analysis and honeypot collection; supported detection and attribution pipelines used by downstream intelligence and defense teams
  - **Recognition.** Our team was awarded the Israel Defense Prize for its contribution to the effort

- **Mobileye** · Software Engineer · Jun 2018–Mar 2019

  Algorithms for a pipeline transforming driving video into centimeter-level HD maps

  Python · NumPy / Pandas · Spark

  - **HD-map algorithms.** Developed geometric and signal-smoothing algorithms to reduce noise in map overlays; built automation for manual-review workflows, improving review speed and map-validation efficiency
  - **Data pipelines.** Solved large-scale graph and geometry problems, optimizing NumPy/Pandas workflows in distributed Spark environments

PUBLICATIONS & PROJECTS

Research Software Developer · D+ / X+

Raviv Research Group · X-ray scattering research

- Designed and implemented C++ algorithms for X-ray scattering analysis in D+ and X+, including modules for polymer structures
- Developed a C# system for identifying repeating subunits in supramolecular structures using geometric transformations, achieving **approximately 10× speedup** at equal computational precision
- Contributed to the design and implementation of an octree-based algorithm for efficient identification of protein hydration shells

Peer-reviewed D+ publication · 2019

A. Ginsburg, T. Ben-Nun, R. Asor, A. Shemesh, L. Fink, **R. Tekoah** et al. *D+: software for high-resolution hierarchical modeling of solution X-ray scattering from complex structures*. *J. Appl. Crystallogr.*, 52:219–242

GraFT – Alignment-Free HGT Detection · github.com/roeetekoah/GraFT · 2025

Built a scalable graph-based system over **~324K proteins** across 50 species, pruning 911K similarity edges to 92K. Detected HGT candidates using k-mer Jaccard similarity, species-pair normalization (median/MAD), and graph-structural anomaly scoring

EDUCATION

The Hebrew University of Jerusalem
B.Sc. in Computer Science
2023–Present · Expected 2027

Gamma Cyber Program (Matzov), IDF
Selective Cyber Defense Program
2019–2020 · GPA 95

OS internals, network security, vulnerability research, reverse engineering, exploit analysis, malware analysis, memory forensics

SELECTED COURSEWORK

Machine Learning	93
Complexity Theory	93
Image Processing	97
Natural Language Processing	100
Advanced Natural Language Processing	
Operating Systems	90
Quantum Computation	
Bio-Algorithms	98
Computational Genomics	98

SKILLS

Programming & Data

C / C++ · Python · C# · Go · IDAPython · NumPy / Pandas · Spark

Algorithms & Research

Graph algorithms · Computational geometry · Bioinformatics · Computational genomics · X-ray scattering

Systems & Platforms

Windows internals · Linux / Android internals · RTOS / embedded · Distributed systems · JTAG / UART / SWD · Firmware extraction · Protocol RE

Reverse Engineering

IDA Pro · Ghidra · Binary Ninja · GDB · Frida · Volatility · Wireshark · ProcMon / ProcDump · Assembly (x86/x64, ARM)

INTERESTS

Algorithms, systems, security research, computational biology, machine learning